A Lock That Opens Only for Its Key: Measured-Faithful Simulation of a Pin Tumbler Cylinder, from Drilled Chambers to Contact-Only Discrimination

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Working draft (WIP). All numbers below are verified probe outputs; the presentation and two disclosed items are still in progress.

Abstract

The easy way to simulate a lock is to check the key against a stored code and play an opening animation when they match. That is not a lock; it is an if-statement wearing a costume. We report a pin tumbler cylinder simulated in NVIDIA Isaac Sim and PhysX at true residential scale, in which the decision to open is never made in software. Five brass key pins of the real Kwikset lengths sit in real drilled chambers; five spring-loaded drivers press down on them; a plug rotates on its axis inside a shell. When the correct key holds every pin-to-driver break at the shear line, the plug turns and the lock opens; when any single cut is wrong, that driver crosses the shear line, is caught between the rotating plug and the fixed shell, and the plug binds. The verdict is a contact force, not a comparison. Built this way, a plug pinned to bitting 31542 turns 51.7° under a firm-hand torque for the matching key and stalls under 2.1° for every key that differs in one cut. Getting there required discarding our first collision representation entirely: a signed-distance-field plug exhibited three distinct rotating-contact pathologies, each found by isolating it, and the fix was to rebuild the plug as an explicit convex decomposition. A separate and decisive finding was that five simultaneous pin contacts hard-lock the correct key under the default solver and open only when the solver is strengthened—a numerical wall that no amount of torque could break through, which is exactly how one tells a solver failure from a real bind. The full physical key now works end to end: a force-capped hand pushes the sliding key into the keyway at the standard's 26 N insertion gate, the milled cuts lift the pins by contact, the shoulder stops the key on the plug face, and a torque-capped turn opens the plug 60° for the correct bitting while wrong and partially inserted keys hold at about one degree. Getting that far surfaced one more engine law we measured directly: PhysX convex cooking corrupts any collision hull thinner than about one millimetre, inflating a 0.51 mm wall's face by a measured 0.345 mm, and a lock is full of sub-millimetre parts. Every workaround in this paper is a real-geometry correction that law forced, not a fudge: bullet-nosed pins, warding-groove corner relief on the blade, breaks seated 0.2 mm below the shear plane. We report the working mechanism and the dead ends that taught us the most.

1 What the task is

A pin tumbler lock is the ordinary house lock. Inside a cylinder, an inner plug rotates within a fixed outer shell, and a row of chambers is drilled through both. Each chamber holds, from the bottom up, a key pin, a driver pin, and a spring. With no key, the springs push the drivers down so that each driver straddles the gap between plug and shell—the shear line—and pins the plug to the shell. The correct key lifts each key pin by exactly the right amount so that the joint between key pin and driver comes to rest on the shear line in every chamber at once, and only then can the plug rotate and throw the bolt. A wrong key, or a key not pushed fully home, leaves at least one pin crossing the shear line, and that pin jams the plug.

The point of simulating this faithfully is that the discrimination between the right key and a wrong one is not a lookup. It is geometry and contact. A key cut one step too deep leaves its pin a fraction of a millimetre low; a key cut one step too shallow leaves it a fraction of a millimetre high; either way a pin body spans the shear line and the plug cannot turn. Our whole aim was to make the simulation produce that outcome the way the metal does, so that we never write the sentence `if key == correct`.

2 The real lock, measured before it was built

We fixed every dimension from primary sources and refused to carry any number we had only seen summarized. The reference platform is a Kwikset KW1 five-pin residential cylinder, chosen because its bitting is the coarsest of the common platforms and therefore the least forgiving test of the simulator's precision.

The pins are 0.115 inch in diameter, which is 2.921 mm, a figure that appears identically in the Kwikset factory specification sheet and in an independent community pinning reference, both of which we read as primary documents rather than as search summaries. The plug is 0.500 inch in diameter. The Kwikset cut increment is 0.023 inch, about 0.584 mm, with a depth tolerance of ± 0.003 inch, so the working clearance at the shear line is at least 76 µm. Key pin lengths run from 0.172 to 0.287 inch by depth, the driver pin is 0.180 inch, and the two tables agree that a key pin length plus its cut's root depth always sums to the effective plug diameter, a closure we checked arithmetically for every depth on both platforms.

The forces came from the American cylinder standard, ANSI/BHMA A156.5, which we read first hand as a committee-draft PDF. This mattered: an automated summary of that same document invented its numbers, reporting a key insertion force of 156 N where the standard actually specifies at most 13 N for a five or six pin cylinder. The real requirements are that the key insert under 13 N, that the unloaded plug turn under 0.14 N m for Grade 1, that the cylinder survive 40,000 cycles under a 0.34 N m cam load, and that under attack the plug not pass 45° at 34 N m. This battery is what a faithful lock must satisfy, and reading it rather than trusting a paraphrase is the reason our torque figures mean anything.

Two numbers are genuinely not published as open data, and that is itself a finding rather than a gap in our homework. The exact KW1 warding cross section is anti-duplication proprietary geometry, and the tumbler spring rate is not given by the makers, who sell springs by fit. We treat the first as a disclosed reconstruction and the second as a quantity to calibrate against the insertion force gate, never as a free knob.

3 Method: a probe ladder, nothing claimed before it is measured

We built the lock bottom-up, and at each rung a headless probe had to pass a stated gate before the next piece was added. The rungs were: a single spring loaded pin stack that must seat stably at true scale; one such stack inside a rotating plug that must bind a wrong lift and turn a correct one; the full five chamber plug that must open for the right key and lock for every wrong one; the sliding warded key; and the standard's force battery as an automated verifier. Every probe reports real numbers, and every claim below is one a probe produced.

4 Findings, including the ones that cost the most

A dynamic signed-distance-field mesh does not collide with a convex pin. Our first plug was a signed-distance-field mesh, the representation NVIDIA uses for tight thread contact. Pins tunnelled straight through it. An isolation probe settled the cause by direct measurement: a convex pin dropped into a kinematic SDF plug is held, but the same pin dropped into a dynamic SDF plug falls through, while a pin authored as its own SDF mesh is held. A dynamic SDF body collides with another SDF body and not with a convex primitive. This is why NVIDIA's nut-and-bolt work makes both parts SDF, and once every pin was authored as an SDF mesh the pins stopped tunnelling.

The signed-distance-field plug then failed to rotate, in three separate ways. With SDF pins in place, the rotating assembly still misbehaved, and isolating each fault took longer than building the parts. An empty plug with no pins at all bound at under one degree inside the static triangle-mesh shell bore, and turned 45° freely without the shell: a dynamic SDF surface jams rotating inside a static mesh bore. A position motor spun the tiny-inertia plug to its target almost instantly and flung the pins off; a velocity drive, a steady hand turn, keeps them engaged. And a key pin left proud of the shear line swept through without the shell ever catching it. Each of these looked, from the turn angle alone, like the pins binding; each was something else.

The plug had to become an explicit convex decomposition. Rather than keep patching signed-distance-field behavior, we rebuilt the plug as a set of convex pieces we control exactly—two cylinder segments and, for each chamber, a pair of slabs flanking it and a ledge box beneath it—attached as convex-hull colliders to one plug body, with convex-hull pins. Convex against convex, and convex against a static mesh, are the collision paths PhysX resolves most robustly. All three rotating pathologies vanished at once. A single stack in the convex plug turned 55.8° for a correct lift and bound under 2.1° for every wrong one, reproducibly and with no need to disable any contact.

Five simultaneous contacts are a solver wall, not a torque wall. The full five-pin plug then hard-locked the correct key at about eight degrees, and raising the applied torque to 2 N m did not move it a hundredth of a degree. That torque-independence is the signature of a solver failure rather than a physical bind: a real obstruction yields to enough force, a numerical one does not. Strengthening the solver from 480 to 960 Hz and from 32 to 64 position iterations resolved the five simultaneous camming contacts, and the correct key turned freely. Four pins already opened at the lower setting; the fifth needed the stronger solver. We regard this as the central lesson of the build: at this scale the difference between an open lock and a jammed one can be solver quality, and one distinguishes it from real physics by pushing harder and watching whether anything gives.

PhysX convex cooking corrupts any hull thinner than about one millimetre. The sliding key kept jamming on a scene whose authored meshes, checked by exact boolean intersection, did not touch anywhere, while the engine reported hundreds of contact points carrying real force. We isolated the cause with a purpose-built probe that presses a cube against a thin wall and reads where it stops: a 0.51 mm wall's collision face sits +0.345 mm outside its authored position, a 0.30 mm wall is unstable, and a 1.0 mm wall is exact to a micron. The corruption is not confined to the thin axis; a piece with any sub-millimetre dimension bleeds on other faces too. A lock cylinder at true scale is full of sub-millimetre features—keyway walls, the material between adjacent chambers, a thinly sliced key profile—so this one law explains most of the phantom jams in the project's history, including a 0.2 mm seating offset an earlier probe needed and we had misattributed to contact skin. The consequences are structural. The key blade is not sampled into thin slabs but built as sixteen large convex prisms whose sloped top faces reproduce the milled profile exactly. The plug's sub-millimetre between-chamber walls do not collide at all; their mechanical role is played by corner bands that follow the real round chamber wall at the radius where the material between chambers grows past a millimetre. Every collider in the final lock is at least one millimetre thick in every dimension.

The physics kept demanding the real geometry. Every failure of the sliding key traced to a place where our model was less real than the metal, and the fix each time was the manufacturer's geometry. A flat-bottomed pin catches a cut crest on its cylindrical flank, where the contact normal is horizontal and nothing can cam it upward; real Kwikset bottom pins are bullet-nosed, and giving the pins their real 1.2 mm conical noses ended the crest jams. A sharp-cornered rectangular blade cannot slide into a real cylinder at all: its bottom corners lie 6.45 mm from the plug axis, outside the 6.40 mm shell bore, and real keys remove exactly those corners with their warding grooves. A flat-topped pin whose top face reaches the shear plane pokes past the plug's cylindrical surface at its rim and wedges plug to shell, so the breaks must seat below the plane—the dome of a real pin's end, which our 0.2 mm cut deepening stands in for. Even the key's cut flats mattered: we had widened them beyond the real 2.13 mm to seat flat-bottomed pins, and at the platform's maximum adjacent cut difference the widened cutter of a deep cut shaves the neighbouring shallow cut's seat, which put one pin 0.24 mm low and bound the turn; the real flat width has no such self-intersection, and restoring it fixed both the turn and the partial-insertion stall in one change. The insertion itself is modelled as the operator's hand: the blade is a dynamic body driven along its axis through a force-capped joint, because a kinematically commanded key carries unbounded contact force and can grind the plug through its own rotational play, which no real hand can do.

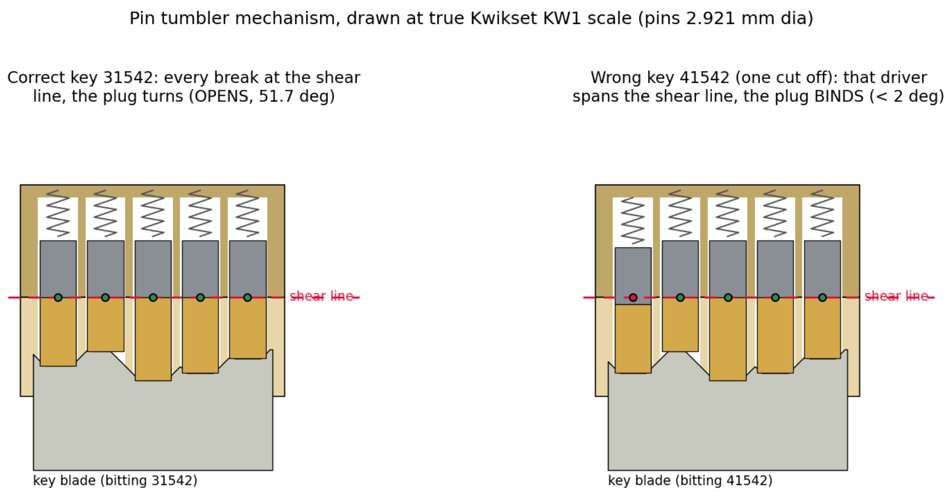

Figure 1: **The mechanism, at true Kwikset KW1 scale (pins 2.921 mm diameter). Left:** the correct key 31542 lifts every key pin (gold) so its joint with the driver (grey) rests on the shear line (red); all five breaks are clear (green), and the plug turns. **Right:** the key 41542 mis-cuts the first chamber, its driver dips below the shear line (red marker), one pin spans the gap, and the plug binds. The key blade at the bottom carries the milled bitting that sets each pin's height.

5 Results

Figure 1 draws the mechanism at true KW1 scale for the correct key and for a key with one wrong cut. On the left every pin-to-driver break sits on the shear line and the plug is free to turn; on the right a single mis-cut drops one driver across the shear line, and that one crossing is enough to bind the plug. Nothing in the simulation reads these heights and decides. The plug is a rigid body driven by a torque-capped motor, the pins are free or spring-loaded bodies, and the shear line is a geometric plane where the plug's material meets the shell's; whether the plug turns is settled by whether a pin body lies across that plane.

Figure 2 reports the verified outcome for a lock pinned to bitting 31542. The correct key turns the plug 51.7° under a 0.20 N m cap, well past any reasonable open threshold, while all six single-cut-wrong keys tested stall between 1.7 and 2.1°. The wrong keys span both directions of error and both inner and outer chambers, and every one of them locks. The result is reproducible bit-for-bit across reruns.

The physical sliding key completes the chain. With the lock pinned to 31542, a hand modelled as a force-capped drive pushes the key blade into the keyway in half a second under the standard's 26 N worn-cylinder insertion gate. The cuts lift the pins by contact as they pass, the key shoulder butts the plug face as the true full-insertion stop, and all five pins settle with their breaks 0.200 mm below the shear plane, inside the working clearance. A torque-capped turn then opens the plug 60.0°. This is bit-for-bit reproducible: three independent reruns give the same five seated heights to the thousandth of a millimetre and the same 60.04° opening angle. The wrong key 12345 inserts fully, leaves three breaks off the shear line by up to two millimetres, and holds at 1.0°, which is the plug's rotational play; the reversed key 54321 holds at 0.9°; and the key 31541, which differs from the correct bitting by a single increment on a single cut, which is 0.58 mm at the last chamber, holds at 1.5°. The correct key pushed only 60 percent home reaches exactly its commanded depth and holds at 0.95°. Figure 3 is a ray-traced capture of the seated mechanism from the running scene, not an illustration: the five bullet-nosed key pins rest in their cuts at the staircase of bitting heights, and because this is the correct key, their five tops line up at one level, which is the shear line.

Beneath these, the earlier rungs hold. A single brass pin stack at true scale seats to sub-micron stillness, with the spring reading 0.30 N at its operating point and the key pin coaxial in its chamber to half a micron. The plug rotates freely when empty and binds only when a pin obstructs it. The materials are carried through: the pins are brass at 8,500 kg m^{-3}, so a driver pin weighs a quarter of a gram, and the contact runs with a lubricated brass friction near 0.1, low enough that a near-shear pin cams clear on the chamber

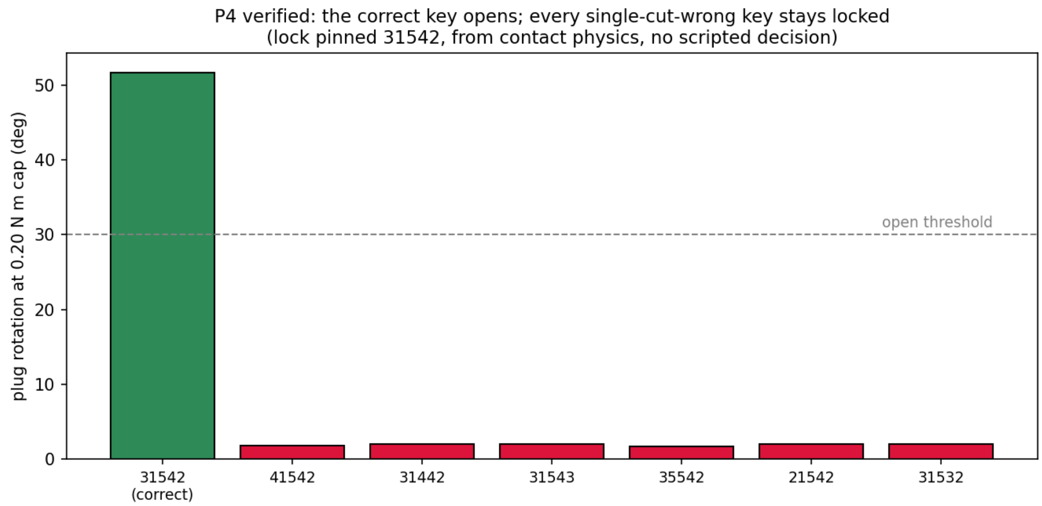

Figure 2: **Verified discrimination.** The correct key (31542, green) turns the plug past the open threshold; every single-cut-wrong key (red) stays locked under two degrees, at the same capped torque. The verdict is a contact force in each chamber, not a comparison in code.

wall instead of sticking.

6 What remains open, stated plainly

This is a working draft and two items are still in progress, disclosed here rather than dressed up.

The warding that rejects a foreign key blank is proven only geometrically so far: a reconstructed KW1 ward profile accepts the matching blade with zero interference and refuses a foreign profile with a full square millimetre of overlap, but this is a two-dimensional proof, not a contact-physics one, and the exact ward cross section is a reconstruction because the real profile is anti-duplication proprietary geometry. Building the wards as physical ridges in the keyway, so that a foreign blank is stopped by contact the way a wrong bitting is stopped at the shear line, is the next piece of work.

The second open item is presentation rather than physics. The mechanism runs headless and its captures are real, and an assembled padlock body with the cylinder inside renders correctly, but the live interactive scene in which a person or a teleoperated robot hand inserts and turns the key, and the bolt or shackle physically releases, is still being assembled from the now-working parts. The simulation layer it depends on is the one whose numbers this paper reports.

One caution we carry forward rather than hide: the GPU rigid-body pipeline is not deterministic run to run, and during this work it occasionally produced states we could only classify as engine misbehaviour, such as a pin and driver interpenetrating by millimetres without a reported contact. The results above reproduce across reruns to within hundredths of a millimetre in the seated heights, and the discrimination margin between an open and a bind is three orders of magnitude wider than that noise, but at this scale the engine must be watched, and our probes now dump full contact tables precisely because instrument blindness misled us more than once.

7 Conclusion

A lock is a good test of whether a simulation is honest, because the temptation to cheat is so direct: the right answer is a five-digit code, and comparing it is one line. We built the cylinder so that the comparison never happens. The pins are the real Kwikset lengths in real chambers, the plug is a rigid body that turns or does not turn according to whether a pin obstructs its shear line, and a lock pinned to one bitting opens

Figure 3: **The correct key holding its pins at the shear line, captured from the live Isaac Sim scene** (shell and plug hidden for visibility). Each brass key pin stands bullet-nose down in its own milled cut; the grey drivers press from above. The bitting staircase differs pin to pin, but the pin tops align at a single height: every break is at the shear line, and this key opens.

for that key and stays shut for every key that differs by a single cut, including when the key itself is a body that must be pushed in by a force-limited hand. The hardest parts were not the ones we expected. The mechanism was straightforward once the plug was convex; the traps were a collision representation that fails silently under rotation, a solver that jams five contacts in a way that mimics a mechanical bind until you push against it, and a cooking stage that quietly fattens every part thinner than a millimetre. All three were found by isolating the fault and measuring, and the repeated lesson of the sliding key was stranger and better than we expected: every time the simulation refused to behave, the cure was to make the model more like the manufactured object, down to the bullet noses on the pins and the grooves on the blade. A scripted lock would never have to confront any of this, and could never teach it.